MAHIEDDINE DELLABANI

Software Architect | High-Performance Java & Distributed Systems

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▸ SUMMARY

Software Architect with 10+ years building high-performance Java applications and scalable distributed systems. Specialized in real-time analytics, in-memory databases, and financial services platforms including market risk, credit risk, and FRTB. I also design and develop modern, responsive websites and web applications. Autonomous, curious, and experienced in multicultural environments. Whether leading a team, diving deep into architecture, or crafting a pixel-perfect website, I build great products for both users and developers.

▸ SKILLS

PROGRAMING

JAVA | JAVASCRIPT / TYPESCRIPT | SQL / MDX | BASH | C / C++

FRAMEWORKS

SPRING CLOUD SLEUTH | SPRING BOOT | EXPO | REACT/NEXT.JS

CLOUD SERVICES

AWS - S3 | AZURE - STORAGE | AZURE - DURABLE FUNCTIONS | GCP - VISION API

SPOKEN LANGUAGES

ENGLISH | FRENCH | ARABIC | SPANISH

▸ EXPERIENCE

Software Architect

04/2025 - PRESENT

Real-Time Analytical Platforms (ActiveViam Accounts) - Full Remote

Leading technical delivery and architecture for real-time risk analytics platforms serving Tier-1 banking clients. Driving design decisions, mentoring a team of 6 engineers, and owning end-to-end technical execution with full autonomy — from migration strategy to production-grade feature delivery.

Santander — Market Risk

🔥 **Direct Query with Databricks:** Owned the full setup of Direct Query from A to Z, enabling the platform to query risk data directly from **Databricks** at scale — covering schema design, SQL dialect adaptation, naming conventions, and production deployment.

🚀 **Platform Migration:** Key driver of the platform migration, delivering it end-to-end while simultaneously implementing new features across **PnL**, **Sensitivity**, and **VaR** cubes.

📈 **Performance & Observability:** Tuned query performance and set up end-to-end observability with **OpenTelemetry**, **Prometheus**, and **Grafana** to ensure reliability at production scale.

BBVA — Risk Viewer

🔥 **Projection Engine:** Leading the design and implementation of the **Rho Basis Projection** feature — a high-priority capability for traders that projects basis swap curves into related curves for sensitivity analysis across multiple currencies.

🚀 **Platform Migration:** Led a cross-framework migration spanning **Spring Boot**, **Spring Batch**, **Atoti**, **JDK 21**, and **AWS libraries** — ensuring continuity for all existing risk calculations and reporting workflows.

⚡ **Performance Optimization:** Driving performance tuning and caching strategies for real-time projection calculations across large portfolios.

ATOTI | DIRECT QUERY | DATABRICKS | MARKET RISK | SPRING BOOT | JAVA 21 | OPENTELEMETRY | KAFKA | AWS | POSTGRESQL | PERFORMANCE | TEAM LEADERSHIP

Freelance Web Designer & Developer

2024 - PRESENT

Self-employed - Full Remote

Designing and developing modern, responsive websites and web applications for businesses and creatives.

🎵 **Love Reaction:** Music label website featuring a custom audio player, playlist management, SoundCloud RSS feed integration, and dark/light theme switching.

🩺 **Collectif Entraide:** Business website for a first aid training organization with session booking, admin dashboard, email notifications, and SEO optimization.

💼 **Personal Portfolio:** This retro-themed portfolio site built with Next.js, Tailwind CSS, and TypeScript.

NEXT.JS | REACT | TYPESCRIPT | TAILWIND CSS | SHADCN/UI | HEROUI | DRIZZLE ORM | TURSO | NEXTAUTH | VERCEL | RESPONSIVE DESIGN | SEO | FRAMER MOTION

R&D Software Architect

09/2023 - 02/2025

Huawei - Full Remote (1 week per month in Grenoble)

Grenoble Fermat Lab. is responsible for advanced technical research and development of Model-based and Formal methods for different business domains including Automotive and ICT. The team is responsible for building formal modeling, simulation, verification and code generation tools for the design and development of trustworthy and efficient embedded software.

🔥 **Scalable & High-Performance DSL Runtime:** Designed, implemented, tested, and benchmarked a Java runtime interpreter and exploration engines for a domain-specific language.

🛠️ **API Design & Extensibility:** Designed a user-friendly API to expose and parameterize runtime internals, enabling seamless configuration, enhanced observability, and greater flexibility for developers.

📦 **Canonical Serialization & Database Integration:** Implemented **canonical serialization** for structured data persistence and seamless interoperability. Designed and optimized data handling for **SQLite** and **Neo4j**, ensuring efficient storage, retrieval, and query performance.

🚀 **Java Migration & Modernization:** Led the migration from Java 8 to Java 21, leveraging new features like **Records, Virtual Threads, and Structured Concurrency** to enhance performance and maintainability.

🔍 **Formal Methods & Verification:** Explored modeling technologies for developing trustworthy software.

🤝 **Cross-Team Collaboration:** Worked closely with **Supply Chain teams** to address performance issues and implement their use cases, ensuring high-quality, robust software solutions.

⚙️ **CI/CD & Infrastructure Modernization:** Updated all **CI/CD scripts**, led the migration of **Jenkins**, and ensured a more efficient and reliable build pipeline.

📖 **Engineering Best Practices:** Established team-wide best practices, including **Architecture Decision Records (ADR)**, improved **repository documentation**, and structured the **wiki** for better knowledge sharing.

API DESING | DSL RUNTIME | FORMAL METHODS | PERFORMANCE | BENCHMARKING | SERIALIZATION | SQLITE | NEO4J | JAVA 21 | ECLIPSE OSGI | C++ | SQL | CYPHER | GTILABCI | JENKINS | DOCKER | SONARQUBE | CONCURRENT SYSTEMS | SIMULATION

Open Contributor

09/2023 - 03/2024

SquashQL - Full Remote

SquashQL is an open-source SQL query engine designed to streamline the process of building multi-dimensional queries. At its core, it acts as a middleware layer that stands between SQL databases and multiple clients or front-end applications.

🔥 **Compiled Queries & API Refinement:** Led a significant restructuring of the query API by introducing CompiledXXX objects, separating query user API from query engine internals, ensuring cleaner code and better maintainability. This redesign simplifies query execution, enforces the SRP principle abd enable the introduction of the vector aggregation feature.

📅 **Date Functions Support:** Added support for **date functions** within the query engine, improving query flexibility and user capabilities.

🛠️ **API Enhancements:** Introduced a public total_count measure, counting the total number of results not considering the query limit.

⚙️ **Query Scope & Resolver Optimization:** Streamlined query scope management by introducing **QueryResolver**, replacing static helper methods with a more flexible and scalable approach.

🔄 **String Removal & Field Optimization:** Spearheaded the initiative to replace all **Strings** in query building with typed **Field** objects, paving the way for future POJO query compilations and improving query resolution consistency.

📈 **Execution Graph Simplification:** Simplified the query execution graph, optimizing performance and making it easier for developers to manage complex queries.

API DESIGN | QUERY OPTIMIZATION | PERFORMANCE | SQL | DATABASE ENGINEERING | DATE FUNCTIONS | JAVA 21 | FIELD OBJECTS | COMPILER | CONCURRENCY

Technical Lead

06/2019 - 09/2023

Activeviam - Full Remote (1 week per month in Paris)

As part of the R&D team, I designed and built Atoti, a real-time decisive data analytics software for financial services.

👨‍🏫 **Team Leadership & Mentoring:** Led a team of 6 engineers, overseeing development, design sessions, and code validation. Provided mentorship, supervised project planning, and coordinated team efforts to ensure timely delivery and adherence to best practices. Owner of the distributed, monitoring and aggregation modules.

💻 **Product Development:** Designed, built, tested, and deployed Atoti Java API capabilities: Aggregation engine, real-time updates, distributed computing, MDX querying.

📊 **Monitoring:** Involved in enhancements and implementations of Atoti Application Performance Monitoring stack: Tracing, metrics, and logs.

🤝 **L2 Support:** Solved performance issues and assisted clients in using the APIs.

⚙️ **JDK Features Prototype:** Supervised the internship and university project focused on maximizing the impact of new JDK capabilities in Atoti Java API (projects Loom and Panama).

IN-MEMORY | DISTRIBUTED SYSTEM | AGGREGATION | CONCURRENCY | MONITORING | JAVA 17 | JAVASCRIPT | ZIPKIN | LOGSTASH | GRAFANA | PROMETHEUS | DOCKER | ATOTI | ACTIVEPIVOT | CIRCLECI | ARTIFACTORY | GITHUBACTIONS | SPRINGBOOT | REST | MDX | OLAP | LEADERSHIP | TEAM MANAGEMENT

R&D Software Engineering

09/2018 - 06/2019

INGIMA - Paris

INGILAB is the Innovation Lab and POC Factory of INGIMA, focused on exploring new possibilities with its motto "explore by doing." As part of INGILAB, I worked on various research studies and proof of concepts in multiple fields such as image processing, machine learning, and embedded systems.

- 📄 **PDF Parsing & Extraction:** Developed and optimized a solution for analyzing nutrition table conformities by parsing and extracting relevant data from PDF documents. This project aimed at automating the comparison and validation of nutrition information to ensure accurate product distribution.
- 🧠 **OCR Project with Google Cloud:** Led the OCR (Optical Character Recognition) project using Google Cloud to extract text from images and scanned documents, improving data accessibility and processing efficiency for distribution clients.

OCR | GOOGLE CLOUD | PDF PARSING | MACHINE LEARNING | IMAGE PROCESSING | DATA EXTRACTION | CARREFOUR

Full Time Researcher - PhD in Computer Science and Mathematics

10/2014 - 09/2018

Verimag - Grenoble

My thesis focused on the design, optimization, and validation of distributed real-time applications for embedded systems. Specifically, it delved into using formal methods to address challenges in distributed real-time environments, such as communication delay, optimization issues, and clock drift.

- ⚙️ **Distributed Real-Time Applications:** Designed and optimized real-time applications for embedded systems using formal methods to solve critical issues like communication delay and clock drift.
- 📊 **Formal Methods & Verification:** Applied formal methods, including timed automata and model-based development, to verify the correctness and performance of the distributed systems.

FORMAL METHODS | MODEL-BASED DEVELOPMENT | TIMED AUTOMATA | DISTRIBUTED SYSTEMS | EMBEDDED SYSTEMS | OPTIMIZATION | VERIFICATION | JAVA | ACCELEO | ECLIPSE

Teaching Fellow

09/2015 - 09/2016

IUT Valence - Université Grenoble Alpes - Valence

- 📊 **Introduction to Excel:** Taught freshmen the fundamentals of Excel, covering data organization, formulas, and basic analytics.
- 🗃️ **Introduction to Relational Databases:** Guided sophomore students through database concepts, including SQL queries, schema design, and normalization.

TEACHING | EXCEL | SQL | RELATIONAL DATABASES | DATA ANALYSIS

▸ EDUCATION

Exchange Student in Computer Engineering

FALL 2014

Iowa State University - USA

ADVANCED COMPUTER ARCHITECTURE | RECONFIGURABLE SYSTEMS | DISTRIBUTED SOFTWARE DEVELOPMENT

Engineering Diploma - Embedded Software and Systems

09/2011 - 09/2014

Grenoble INP PHELMA/ENSIMAG - France

- 🚀 **Engineering thesis:** Vectorization of compression algorithms using SIMD instructions. Realtors: Prof. S. Viardot, Ing. R. Schulze, Dr. T. Willhalm. Thesis activity carried out during the final year project at SAP SE, Walldorf, Germany.

MATHEMATICS | PROGRAMMING | OPERATIONAL RESEARCH | OPERATING SYSTEMS | REAL-TIME EMBEDDED SYSTEMS | HARDWARE DESIGN | C++ | SAP HANA